

(a) Event Log Structure

Case ID	Activity	Timestamp	Resource
1	ER Registration	09:00	Nurse A
1	ER Triage	09:15	Nurse A
1	ER Sepsis Triage	09:30	Dr. X
1	IV Antibiotics	09:45	Dr. X
1	Admission IC	10:00	Dr. X
2	ER Registration	10:30	Nurse B
2	ER Triage	10:45	Nurse B
2	IV Liquid	11:00	Dr. Y
2	Admission NC	11:15	Dr. Y
3	ER Registration	08:00	Nurse C

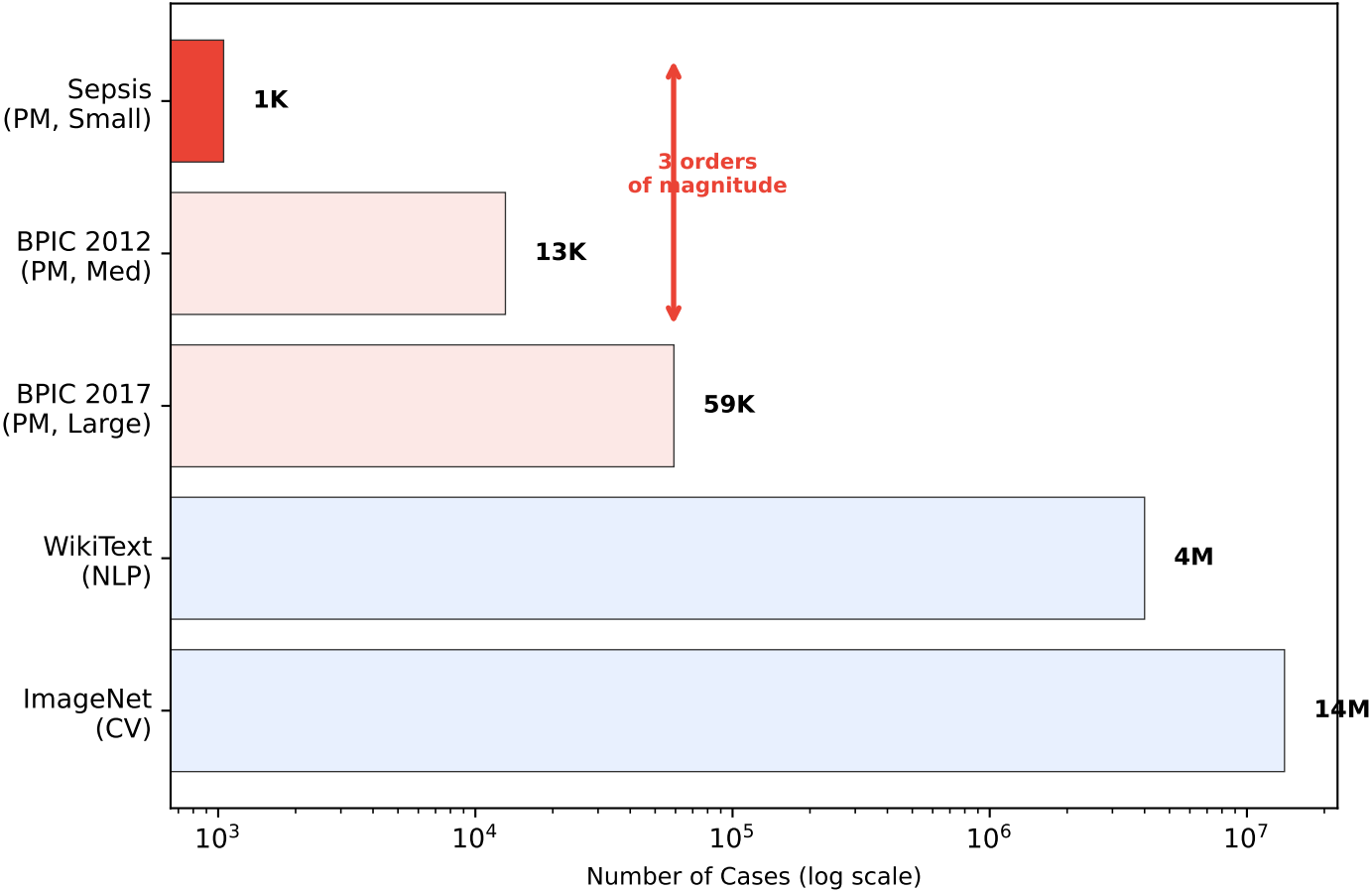
Each row = 1 event. Each trace = sequence of events for 1 patient.

(b) Trace Variants in Sepsis Log

t1	ER Reg.	Triage	Sepsis Triage	IV Liq.	Adm. NC	CRP	Rel. A
t2	ER Reg.	Triage	Sepsis Triage	IV Abx.	Adm. IC	Lac. Acid	Rel. C
t3	ER Reg.	Triage	Sepsis Triage	IV Liq.	Adm. IC	Leuc.	Rel. E
t4	ER Reg.	Triage	Sepsis Triage	IV Abx.	Adm. NC	CRP	Rel. B

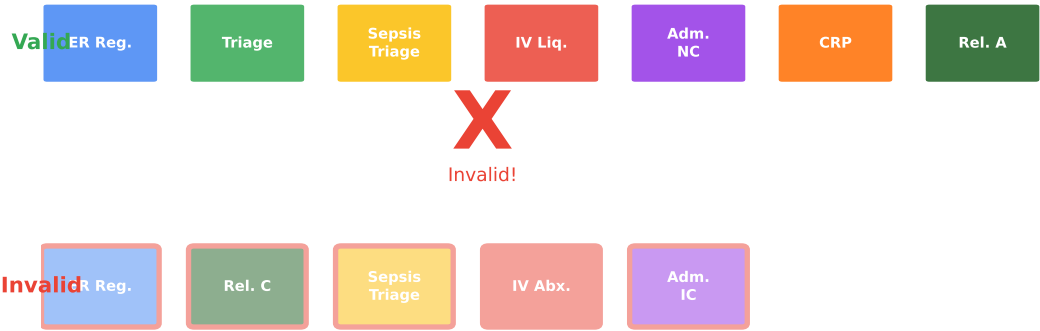
830 variants from only 1,050 cases = high variability

(c) Data Scarcity: Sepsis vs Typical DL Dataset



Sepsis has 1,000x fewer cases than typical DL training sets

(d) The Augmentation Challenge



Random augmentation breaks process semantics:

"Release" cannot happen before "Admission" + "Tests"

Our Solution: Multi-LLM Consensus

Generate with reasoning + Verify with 3 independent checks